

Sayan Banerjee

Worcester, MA | 774-519-9984 | SaBanerjee@clarku.edu | [LinkedIn](#)

EDUCATION

Master of Science in Information Technologies; Concentration in Cyber Security (STEM) May 2024
Clark University; Worcester, MA

Bachelor of Technology in Computer Science and Engineering May 2020
University of Engineering and Management (UEM), Kolkata

EXPERIENCE

Clark University; Worcester, MA January 2024 - Present
Teaching Assistant

- Graded assignments for 30 students with specified rubrics for the Spring 2024 Cybersecurity Fundamental course, ensuring consistent and fair evaluation, contributing to an overall improvement in student performance.
- Collaborated with the instructor weekly on improving the course curriculum, setting question papers for exams, and providing valuable student-centered insights to enhance the educational experience.

Wipro; Kolkata, India September 2020 - December 2022
Network Test Engineer

- Led a pivotal project for Cisco by conducting comprehensive server-side application tests, proactively resolving issues, and contributing to generating over \$300 million in annual partnership revenue.
- Managed seamless deployment of multiple Web Security appliance builds by conducting system requirement analysis, bug resolution, performance assessment, scalability tests, and stability testing for successful roll-outs.
- Introduced an efficient approach and cleared backlogs, reducing bug escalation, feature testing, and build deployment time by 50%.
- Developed automated test cases and fostered effective collaboration with cross-functional teams to expedite issue resolution within the project, reducing testing time by 40%.
- Improved stakeholder comprehension by 20% through the documentation of test procedures, enhancing user-friendliness.

National Institute of Vocational Training, Kolkata, India January 2020 - June 2020
Data Analyst Intern

- Facilitated fraud detection by leveraging the Hadoop distributed file system to visualize event logs, enhanced data insights, and more effective fraud prevention.
- Achieved a 10% reduction in fraudulent activities by deploying ML algorithms and implementing data visualization techniques.

SKILLS

Languages: Java, Python, C++, SQL, JavaScript, CLI, HTML/CSS, Shell Scripting, XML.

Tools & Framework: Cisco Web Security Appliances, MySQL, OOP, REST API, Splunk, Spring MVC, Maven, Wireshark, Network Protocols, Hadoop, Power BI basics, SIEM, MITRE, Cyber Kill Chain, Kerberos, Jira, OSI model, VPC, Malware Analysis, VLANs, Yara, DNS, DHCP, File Hash Analysis, urlscan, subnet, MS Suite.

Others: Network Testing, Unit Testing, Functional testing, Manual Testing, Quality assurance, Regression testing, Agile.

PROJECTS

The General Data Protection Regulation (GDPR) project for Clark University 2024

- Led an ongoing project to ensure compliance with the GDPR for Clark University's website.
- Identified major vulnerabilities in the cookie management system and optimized security protocols through a comprehensive analysis of over 50 different policy types, safeguarding individuals' privacy rights.

Heart Disease Prediction System - [Hadoop, Python, ML Algorithm]

2020

- Developed a web application empowering users to input daily habits, utilizing a linear regression model to predict the probability of experiencing a heart attack.
- Impacting 5 million potential individuals in India by enabling rapid identification of their risk of heart attacks.

EXTRA CURRICULAR & CERTIFICATIONS

CompTIA Security+ Certification

March 2024

CompTIA.org

Skills: Security protocols (TLS/SSL, IPsec, SSH, HTTPS), Risk assessment and vulnerability management, Incident response coordination, Compliance (GDPR, HIPAA, PCI DSS, NIST), Secure network design, Cryptography (AES, RSA, HMAC).

Boston Consulting Group GenAI Job Simulation on Forage

April 2024

- Developed AI-powered financial chat-bot utilizing Python programming and pandas for data manipulation, integrating complex financial data from 10-K and 10-Q reports to provide user-friendly insights for BCG's GenAI Consulting team.

JPMorgan Chase & Co. Cybersecurity Job Simulation on Forage

March 2024

- Identified that fraudulent activities are predominantly observed in CASH_OUT and TRANSFER transactions, enabling management to prioritize manual reviews on these specific transaction types for enhanced fraud detection and mitigation strategies.

Mastercard Cybersecurity virtual experience program on Forage

March 2024

- Analyzed and identified which areas of the business needed more robust security training and implemented training courses and procedures for those teams, and helped identify and report security threats such as phishing.

Cognizant Agile Methodology Job Simulation on Forage

March 2024

- Applied knowledge of key aspects of Agile methodology by identifying user stories for an innovative interplanetary video-conferencing application and creating a document detailing Scrum role assignments and ceremonies.

Boston Consulting Group Strategy Consulting Job Simulation on Forage

February 2024

- Implemented strategic measures resulting in a 20% efficiency enhancement for a local telecommunications company grappling with declining profits, leveraging comprehensive market research, Excel modeling, and concise stakeholder presentations.

SOC Level 1 on TryHackMe

December 2023

- Acquired expertise in implementing cybersecurity frameworks and policies to establish robust security postures, contributing to enhanced defensive strategies and risk mitigation.
- Configured and maintained security tools to ensure optimal performance and accuracy in threat detection and response, contributing to the overall effectiveness of the SOC infrastructure.

Business Analysis by Microsoft and LinkedIn

August 2023

- Mastered various software tools crucial for business analysis, demonstrating proficiency in utilizing technology to streamline analytical tasks.
- Applied several requirements gathering and analysis techniques, showcasing my ability to effectively gather and analyze business requirements for informed decision-making.